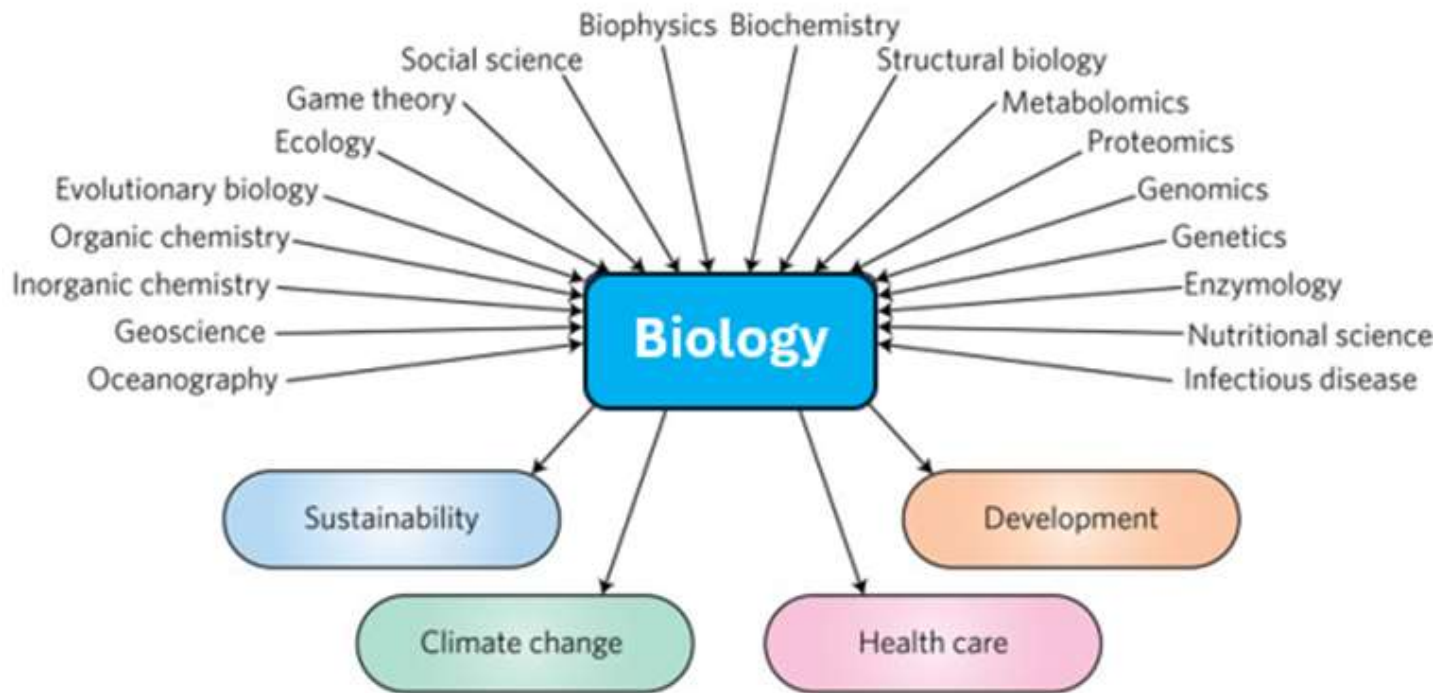

BIOLOGY IS AN INTERDISCIPLINARY STUDY



- Interdisciplinarity in biology refers to combining knowledge and methods from various scientific fields to study biological systems and address complex problems.

WHY STUDY BIOLOGY?



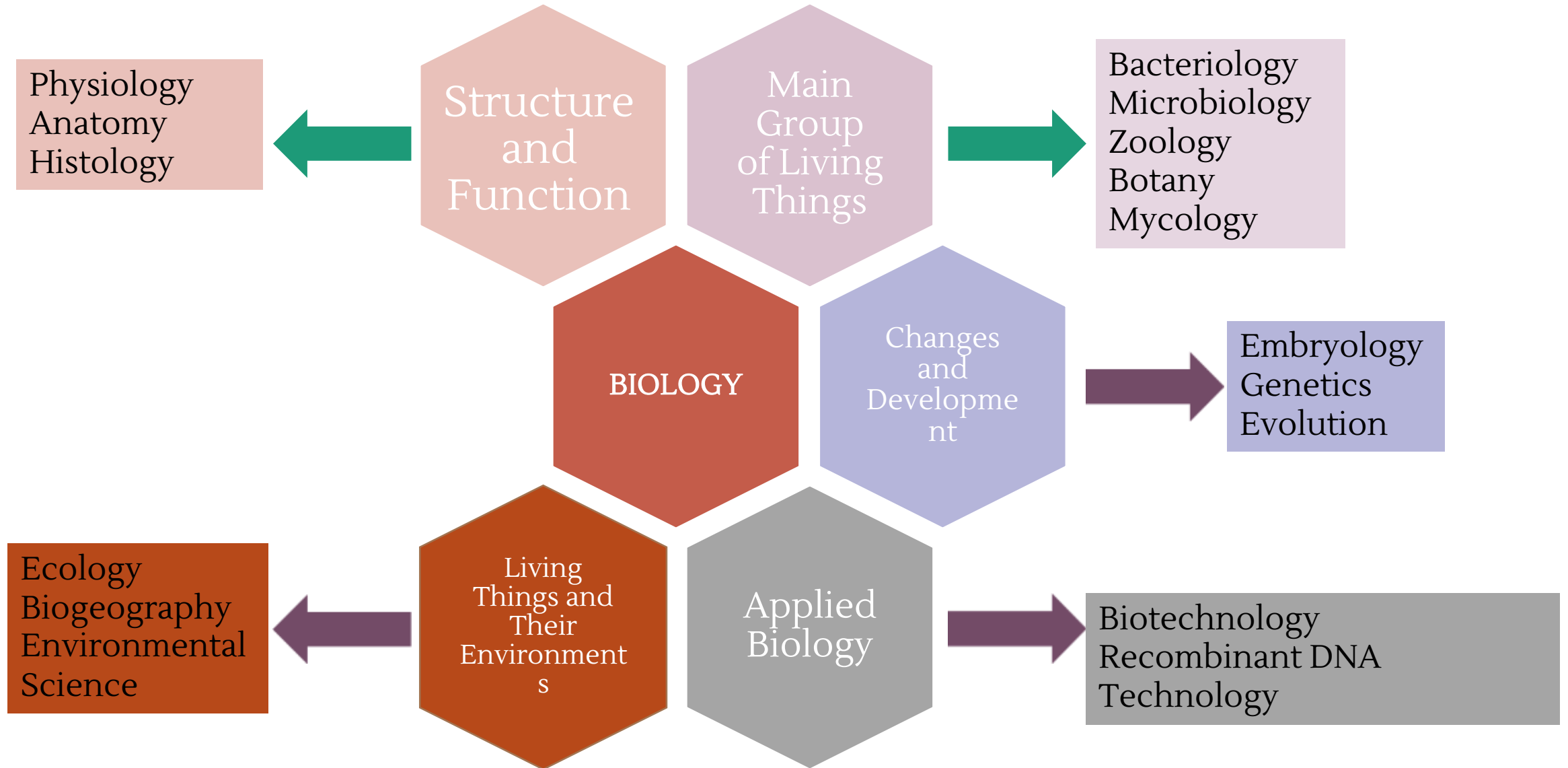
Biology is crucial for understanding the causes, mechanisms, and treatments of diseases.



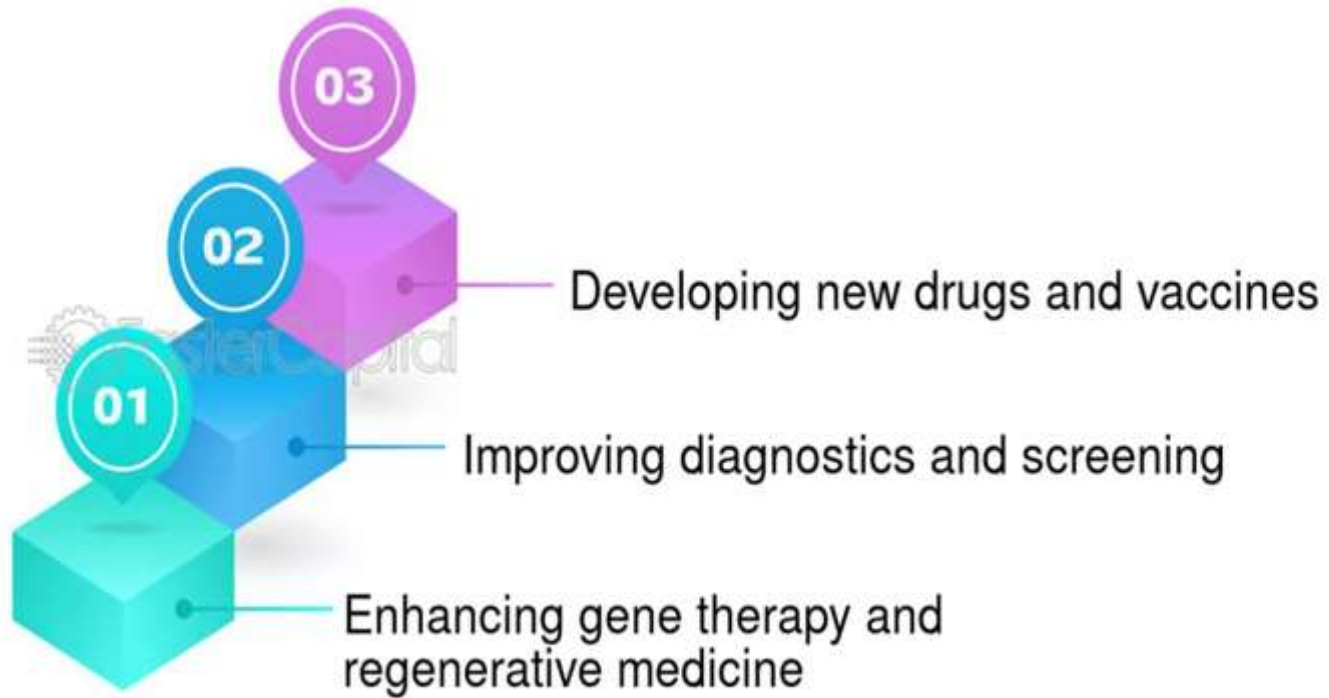
Biology informs our understanding of evolution, biodiversity, and the interconnectedness of life on Earth.



It provides the foundation for numerous careers and drives innovations in medicine, food production, and sustainable practices.



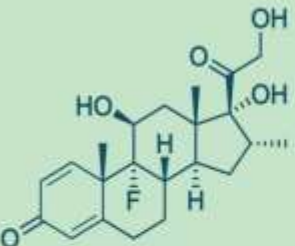
MEDICINE AND HEALTHCARE



USING BIOLOGY TO FIGHT COVID- 19

Using biology to fight COVID-19

Social restrictions are and will continue to be essential to controlling COVID-19, saving lives, and restoring economic growth. At the same time, researchers are using their knowledge of biology to develop new means to control the pandemic. Three large focus areas are:

Small molecules	Antibodies	Vaccines
		
Inhibit viral and human process that enable infection. Activate human processes that fight infection.	Bind to the virus to prevent infection and activate immune responses.	Elicit long term immunity to SARS-CoV-2 in healthy people. Prevent further spread of the virus.

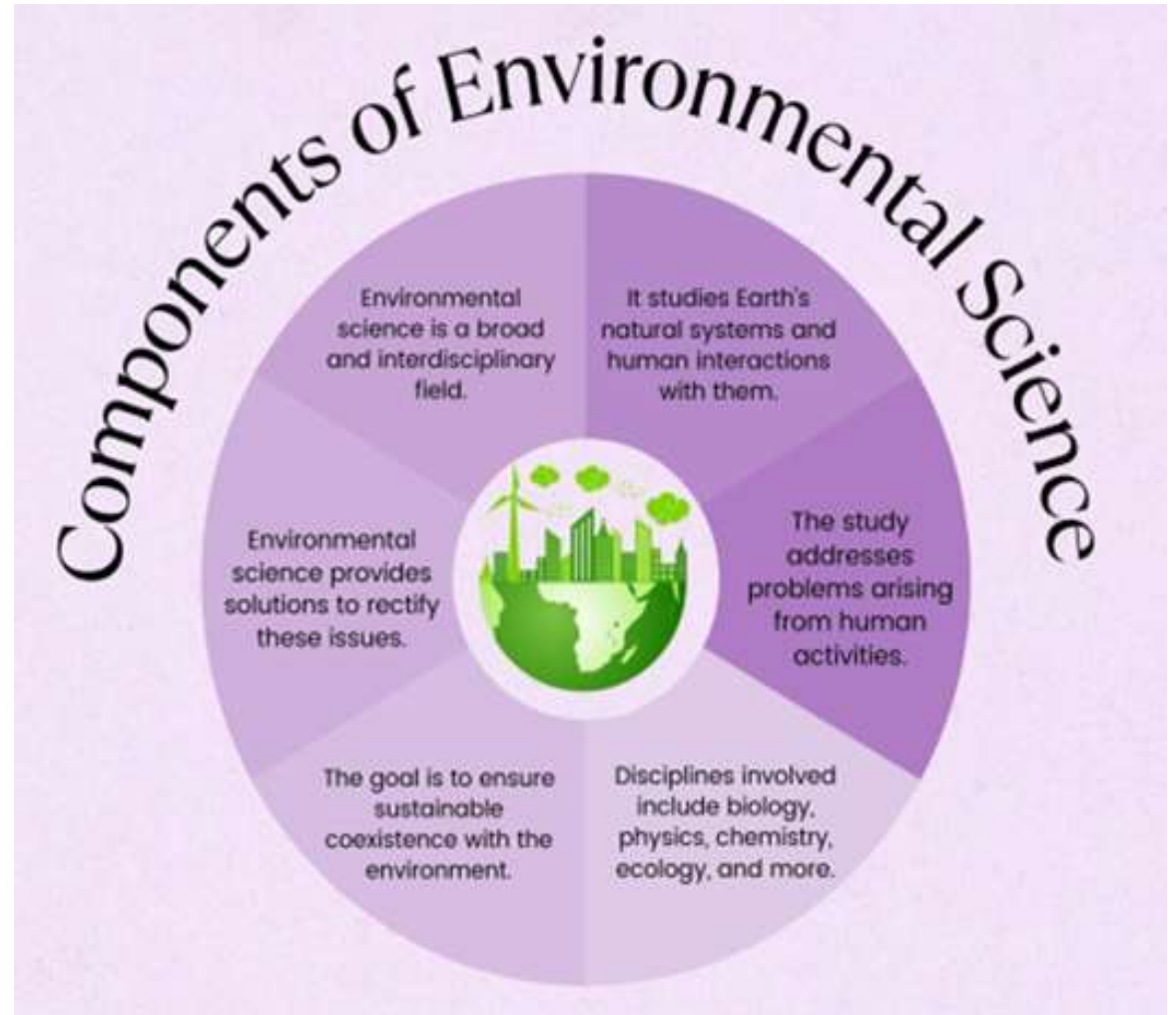
[Reference](#)



AGRICULTURAL SCIENCE

- 1 Increased Crop Yields
- 2 Reduced Environmental Impact
- 3 Improved Resource Efficiency
- 4 Enhanced Crop Quality
- 5 Adaptation to Climate Change

ENVIRONMENTAL SCIENCE AND CONSERVATION



FORENSIC SCIENCE



BIOTECHNOLOGY



Red
medical and
pharmaceutical



White or gray
industrial



Green
agricultural



Gold
bioinformatics
and data



Blue
marine and
aquatic
environments



Yellow
food
production

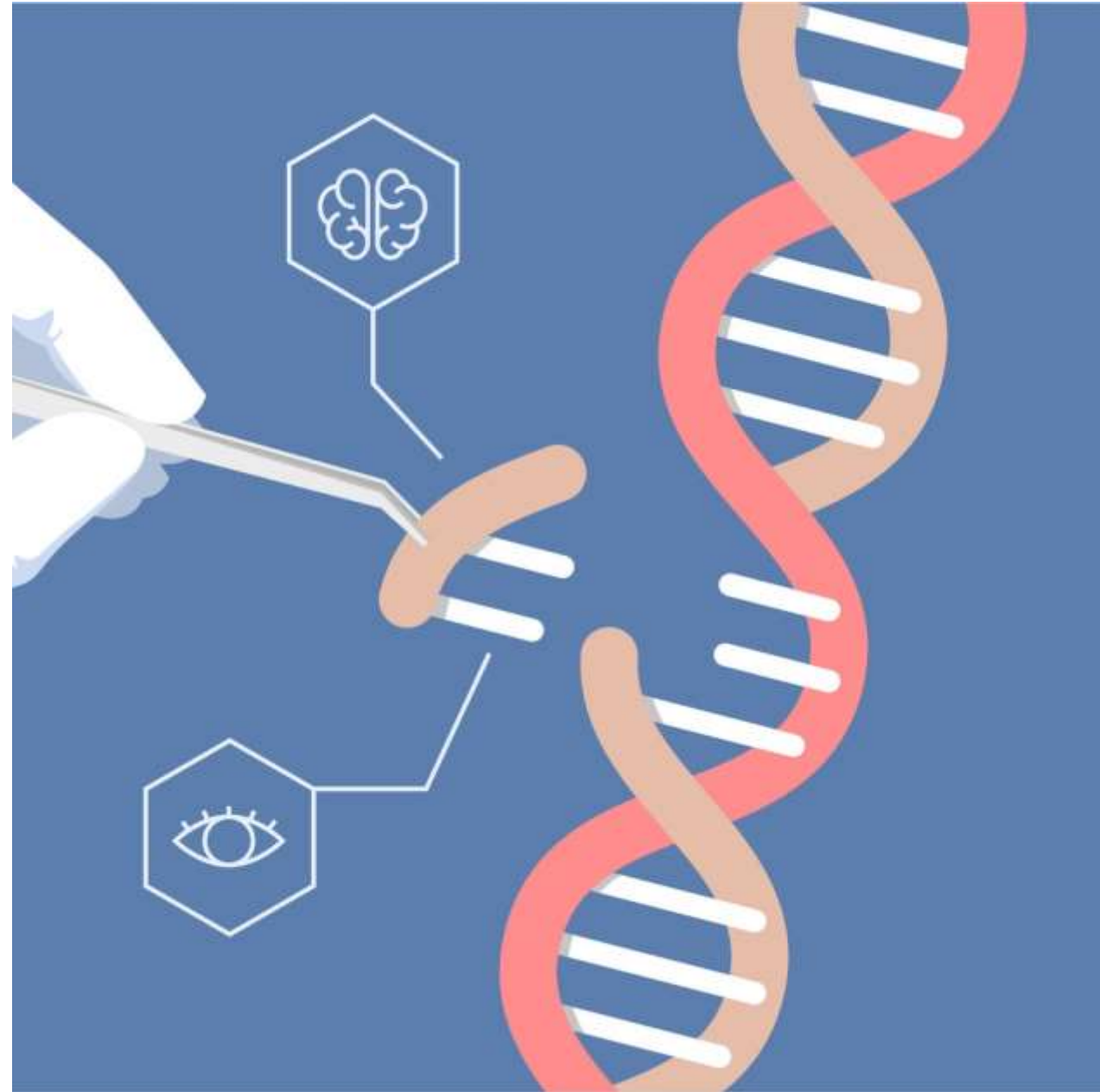


Violet
governance
of ethical
considerations



Dark
warfare

GENE EDITING/ CRISPR CAS



NANOBIOTECHNOLOGY



INDUSTRIAL BIOTECHNOLOGY





ASTROBIOLOGY

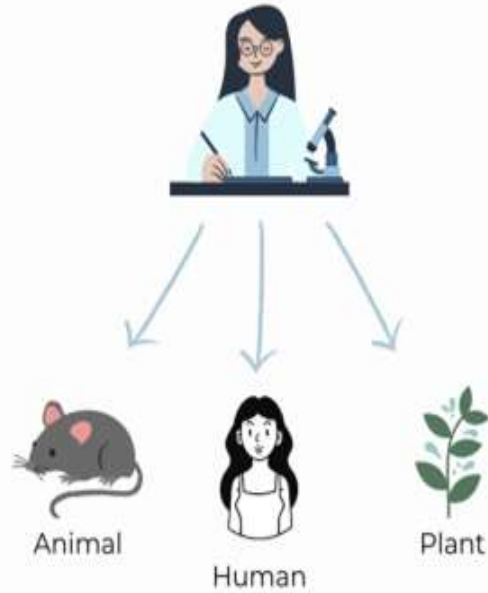
How does life begin and evolve?

Is there life beyond earth? How to work on it?

What is the future of life on earth and in the universe?

BIOETHICS

Subjects of bioethic questions



4 bioethics principles

Beneficence

Non-maleficence

Autonomy

Justice

Bioethical Questions

- **Informed consent:** Ensuring patients understand and agree to medical procedures.
- **Gene editing:** The ethical implications of altering human genes.
- **Artificial intelligence in healthcare:** How AI can be used ethically and responsibly in medical decision-making.
- **End-of-life care:** Ethical considerations surrounding palliative care, assisted dying, and organ donation.
- **Research ethics:** Ensuring ethical conduct in biomedical research involving human subjects or animals.
- **Environmental ethics:** Considering the ethical implications of human activities on the environment and non-human life.

